Unit 1

Introduction To Operational Research (OR)

* Introduction

→ OR is a scientific method of solving complex decision-making problems.

- The main goal of OR is to find the most efficient and effective way to allocate resources, manage systems and achieve objectives.
- It helps in choosing the best possible solution among many options, espically when resources (time, money, materials, labor, etc) are limited.

Why OR is important?

- Improves efficiency and productivity.
- Reduces cost and waste
- Enhance decision making under uncertainty
- Supports strategic planning and policy design.

Example: A company want to minimize shipping cost while delivering products to customers.

- OR Approach
- model the problem as a transportation problem
- Use algorithms to determine the optional routes & shipments.
- Result: Cost saving and faster delivery times.

* History of OR

→ The term OR originated during world-war II when U.S.A. and Britain's Armed forces sought the assistance of scientists to solve complex and very strategical and tactical problems of warfare, like making mines harmless or increasing the efficiency of antisubmarine aried warfare, etc.

Date: ....
Page: ....

* Stages of Development of OR

1, Problem defination
→ Define problems, objectives and constraints.

2, Model Deployment/formulation
→ Create a mathematical representation of the system. (e.g. equations, graphs)

3, Data collection
→ Gather the relevant data to parameterize the model.

4, Solution Genereation
→ Apply appropriate techniques (e.g. simplex method, simulation) to find optimal soution

5, Validation & testing
→ Check if the model accuratory represents reality and refine as needed.

6, Implementation & monitoring
→ Apply the solution in the real world and monitor results & make adjustments as needed.

* Relationship betwn Managers and OR Specialist

Manager: Recognize the problems of the organization
Both: Analyze the problem
ORS: Investigate methods for solving the problems
ORS: Find the optimal solution of the problems
Both: Determine the most effective solution
Manager: Choose the solution to be used
OR: Ensure model accuracy and validity
Manager: Implement the choosen solution in real-world.
Both: Monitor outcomes post implementation

Date:
Page:

* Tools and Techniques of OR

- Linear programming : mathematical technique to optimize a linear objective function under constraints.
- Transportation Method : optimize transportation of goods from sources to destinations at minimum cost.
- Assignment Method : Assign tasks or resources to agents to minimize cost or maximize efficency.
- Queuing Theory : Analyze waiting lines to reduce service time and costs.
- Inventory control : Manage stock levels efficiently to avoid overstocking or shortages
- Replacement Theory : Determine a time at which machine should be replaced to minimize cost & maximize efficency
- Game theory : Find optimal strategies for players in situations of competition or conflict.

* Application of OR

- Manufacturing : production scheduling, quality control and resource allocation.
- Transportation : Route planning for airlines, delivery vehicles, or public transit to minimize cost.
- Healthcare : scheduling staff, managing patient flow or allocating medical resources.
- Defence : Mission planning and resource deployment.
- Retail : Maintaining optimal stock level of items to balance demand and supply
- Business : optimal resource allocation (eg. manpower, materials) to minimize cost & maximize profit.
- Finance & banking : portfolio optimization, risk management, fraud detection and queue management at counter
- Sports : Team selection, game strategy, performance analysis
- Government : policy decision - making and budgeting.

* Limitations of OR

- Data dependency : wrong data gives wrong information.
- complexity : Difficult to understand and solve.
- Time : Time consuming to build models
- cost : Require high cost for specialized software & expert teams.
- simplyfying Assumption: model often assume linearity or certinaty, which may not reflect real situations.
- Resistance to change.